A Unified Vibrational Information Framework: Statistical Bias, Scale Invariance, and the Emergence of Physical Law

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Abstract

This work develops an advanced theoretical synthesis unifying the Vibrational Information Field Theory (VIFT) with Spacetime Vortex Gravity (SVG), the Cosmogenesis Hypothesis (including the Anas Limit), the Hidden Collapse Hypothesis, and observed statistical regularities such as Benford's Law. We argue that these phenomena arise from a deeper principle: reality is governed by a constrained, scale-invariant vibrational information field. Apparent randomness emerges from local decoherence, while global statistical bias reflects structural information flow. The framework reconciles quantum indeterminacy with large-scale order and provides a non-deterministic yet non-random interpretation of physical law.

Preface and Motivation

Physics has historically oscillated between two extremes: strict determinism and fundamental randomness. Classical mechanics implied a clockwork universe, while quantum mechanics introduced irreducible probabilistic behavior. However, modern observations reveal persistent order beneath apparent chaos: scaling laws, power-law distributions, symmetry breaking patterns, and logarithmic statistical biases.

This paper is motivated by a central question: why does randomness repeatedly organize itself into biased, scale-invariant structures? Rather than invoking coincidence, we propose that such behavior is a natural consequence of an underlying vibrational information field.

1 Foundations of Vibrational Information Field Theory (VIFT)

VIFT postulates that the most fundamental entity in nature is information structured dynamically as vibration. Formally, we define an information field $\mathcal{I}(x,t)$ whose states correspond to distinguishable configurations of physical reality.

Each physical system corresponds to a set of allowed vibrational eigenmodes:

$$\mathcal{I}(x,t) = \sum_n A_n e^{i(\omega_n t - k_n x)}. \tag{1}$$

Stability of physical laws arises from resonance selection: only vibrational modes that satisfy global consistency constraints persist.

2 Information Flow, Entropy, and Scale Invariance

Information propagation in VIFT is multiplicative rather than additive. Consider an information measure I evolving across scales:

$$\frac{dI}{d\ln s} = \alpha, \tag{2}$$

where s is the scale parameter. Integrating yields logarithmic dependence, naturally producing scale-free behavior.

Shannon entropy $S = -\sum p_i \ln p_i$ is reinterpreted as a measure of vibrational mode accessibility. Entropy maximization occurs under vibrational constraints, not unconstrained randomness.

3 Benford's Law as an Information-Theoretic Signature

Benford's Law,

$$P(d) = \log_{10}\left(1 + \frac{1}{d}\right), \tag{3}$$

emerges naturally from logarithmic information propagation. Within VIFT, this law is interpreted as a projection of scale-invariant information flow onto numerical representations.

Thus, Benford bias does not imply determinism, but reflects the topology of the information field.

4 Spacetime Vortex Gravity (SVG) within VIFT

SVG proposes that gravitational phenomena arise from rotational dynamics in spacetime. In VIFT terms, spacetime curvature is an emergent effect of phase gradients in $\mathcal{I}(x,t)$.

Define the informational vorticity:

$$\vec{\Omega} = \nabla \times \nabla\phi, \tag{4}$$

where ϕ is the phase of the information field. Gravitational attraction corresponds to synchronization of neighboring vortical modes.

5 Cosmogenesis Hypothesis and the Anas Limit

The Cosmogenesis Hypothesis proposes universe formation via hypermassive collapse exceeding a critical informational density. The Anas Limit is given by:

$$M_{\text{Anas}} = \gamma \frac{c^3}{G^{3/2}\sqrt{\rho_{\text{crit}}}}, \tag{5}$$

In VIFT, this represents a critical point where vibrational modes can no longer remain coherent, forcing a global reconfiguration interpreted as a Big Bang–like event.

6 Hidden Collapse Hypothesis and Inter-Universal Information

The Hidden Collapse Hypothesis extends VIFT beyond a single universe. Multiple spacetime domains coexist within a shared information field. Collapse corresponds to large-scale decoherence across vibrational sectors:

$$\mathcal{I}_{\text{total}} = \sum_U \mathcal{I}_U. \tag{6}$$

Information leakage or coupling between universes explains recurring structural similarities and statistical laws.

7 Randomness, Determinism, and Constraint Realism

We introduce *constraint realism*:

> The universe does not pre-write events, but it pre-structures probabilities.

Quantum randomness arises from local phase uncertainty, while global order emerges from vibrational boundary conditions.

8 Philosophical Implications and Einstein's Intuition

Einstein's objection to fundamental randomness is reinterpreted. God does not play dice without rules; the dice themselves are shaped by informational resonance.

9 Predictions and Future Directions

This framework predicts:

- Universal appearance of scale-biased statistics
- Rotational informational signatures in gravity
- Cosmological constants linked to stability of vibrational modes

10 Conclusion

We have constructed an advanced, unified theoretical framework linking statistical bias, gravity, and cosmogenesis through vibrational information dynamics. The universe is neither chaotic nor scripted, but resonantly biased toward stable informational structures.

This paper serves as a foundational step toward a mathematically rigorous theory of informational physics.